

Shreyaa Sharma

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Software engineer specializing in distributed systems, real-time OLAP infrastructure, and performance engineering. Over the past 3 years at Uber, I've built core components powering Apache Pinot, from query routing and caching to authorization, reliability, and production performance, supporting systems that serve 200M+ queries every day. I care about building systems that are simple, reliable, and efficient at scale.

EXPERIENCE

Software Engineer II – Pinot | Uber India

June 2023 – Present

- Built core infrastructure (routing, caching, client libraries) for Cellar, a distributed Apache Pinot query proxy, contributing to its Tier-1 readiness serving 100M+ queries/day.
- Led ARM benchmarking and JRE17 migration across production Pinot clusters, achieving 20-33% cost reduction per vCPU, 70-90% query latency improvement, and 80% GC reduction.
- Improved reliability and feature capability of Neutrino, a Tier-0 Pinot proxy serving 200M+ queries/day, by extending UDFs, resolving critical production incidents, and strengthening deployment safety.
- Designed and shipped a pluggable L7 authorization framework for the Pinot control plane, integrating with Uber's internal AuthZ infrastructure to enforce request-level access control across the real-time analytics stack.
- Contributed to Apache Pinot OSS: query short-circuiting and API fanout performance optimizations, reliability fixes (MSE join feature flag, bounded thread pool), and security enhancements (Minion AuthProvider, expanded access-control coverage).
- Served as incident commander for 2 high-severity (L4+) production incidents on Tier-0 Pinot infrastructure, driving cross-team mitigation and reducing time-to-resolution.
- Mentored a software engineering intern over a 6-month rotation, guiding project scoping, code review, and technical ramp-up on Pinot infrastructure.

Software Engineering Intern | Uber India

May 2022 – July 2022

- Built a feature release manager for Apache Spark version upgrades, automating rollout workflows and post-rollout observability, reducing developer dependence.

Software Engineering Intern (UberSTAR) | Uber India

May 2021 – June 2021

- Developed an A/B testing suite for Apache Spark-based services in Java, automating workflow testing and enabling early detection of issues in production data pipelines.

PUBLICATIONS

- ["Rebuilding Uber's Apache Pinot™ Query Architecture"](#) – Uber Engineering Blog, Oct 2025
- ["Serving Millions of Apache Pinot™ Queries with Neutrino"](#) – Uber Engineering Blog, Dec 2024

TECHNICAL SKILLS

Languages: Java, Go, Python, C++, Rust, SQL

Technologies: Apache Pinot, Apache Spark, Apache Kafka, Apache Flink, Kubernetes, gRPC, distributed query engines

Areas of Focus: Distributed Systems, Real-time Data Processing, OLAP, Consensus & Replication, Query Optimization

EDUCATION

IIT Roorkee – B.Tech, Civil Engineering

July 2019 – May 2023

- Self taught developer; Transitioned into software engineering during degree via open-source and internship work (SDSLabs, Google Summer of Code, Outreachy).

OPEN SOURCE & COMMUNITY

- Apache Pinot OSS – Active Contributor
- Google Summer of Code (Ceph, 2021) – Integrated Redmine-based issue tracking into the Ceph dashboard for streamlined bug reporting.
- Outreachy (Public Lab, 2020) – Revamped the Public Lab website and productionized a Jest + Puppeteer E2E test suite.
- Software Development Section, IIT Roorkee (SDSLabs) – Developer, Jan 2020 – May 2023.